

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	138781	46.080619
1	[Git, Change, Create]	44929	14.918153
2	[Git, Change, Delete]	22781	7.564166
3	[Java, ByteCode, Method, Member]	13472	4.473221
4	[Java, ByteCode, Parameter]	13306	4.418103
5	[Git, Commit]	11390	3.781917
6	[File, Git]	8286	2.751270
7	[Java, ByteCode, ParameterizedType, Bound]	7286	2.419232
8	[Java, ByteCode, Bound]	7245	2.405618
9	[Git, Change, Rename]	7018	2.330245
10	[Java, ByteCode, Member, Field]	3602	1.196002
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979513
12	[Java, Value, ByteCode, Annotation]	2936	0.974865
13	[Xml, Element]	2162	0.717867
14	[Java, ByteCode, Constructor, Method, Member]	2120	0.703921
15	[Xml, Text]	1450	0.481456
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.368895
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321413
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295514
19	[Type, File, Java, Class, ByteCode]	845	0.280572
20	[Json, Key]	739	0.245376
21	[Value, Json, Scalar]	723	0.240064
22	[Java, Value, ByteCode, Primitive]	673	0.223462
23	[Type, File, Java, ByteCode, JavaType]	649	0.215493
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.191918
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136800
26	[Git, Change, Copy]	298	0.098947
27	[Author, Git, Person]	295	0.097951
28	[Value, Array]	289	0.095959
29	[Committer, Git, Person]	242	0.080353

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

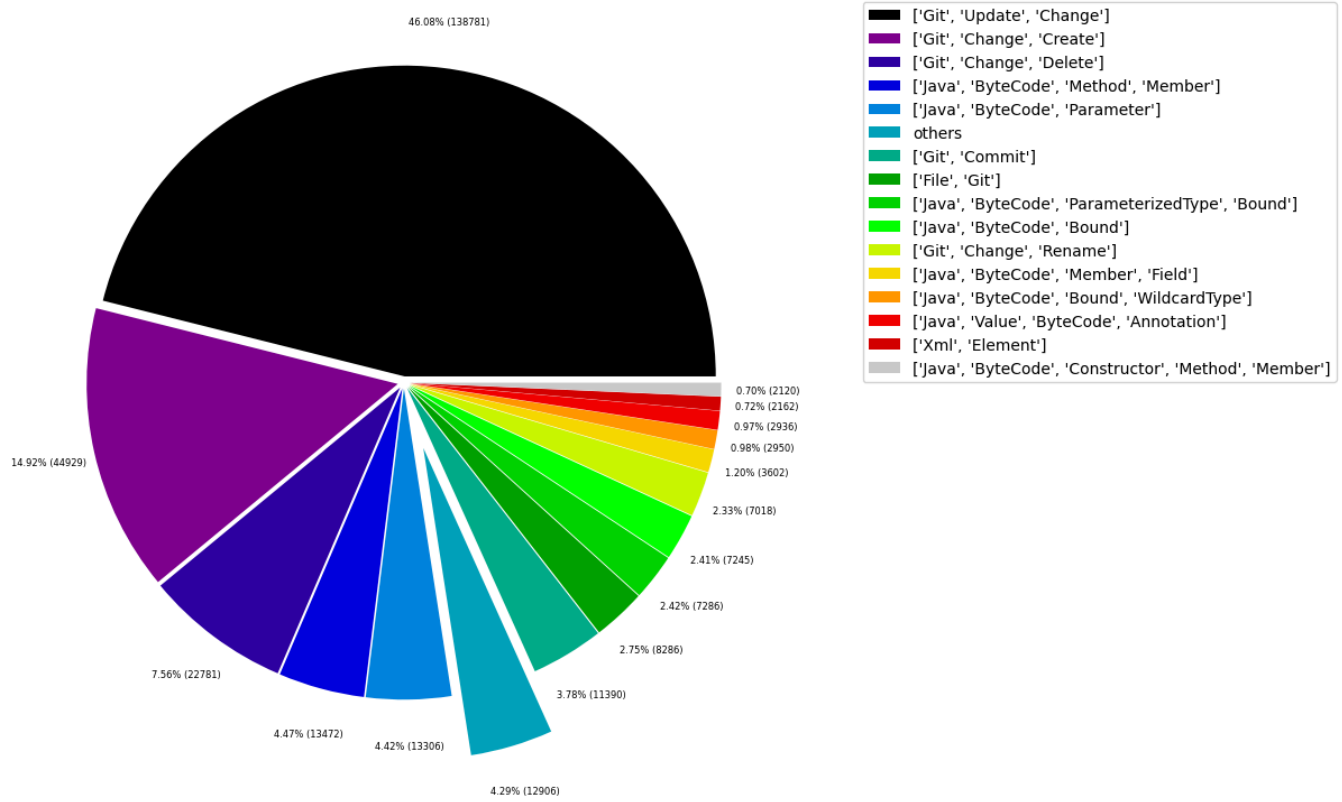


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Array, Json]	6	0.001992
8	[File, Maven, Xml, Pom, Document]	9	0.002988
9	[Type, File, Java, ByteCode, Void]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005313
16	[Maven, PluginExecution]	16	0.005313
17	[Xml, Attribute]	18	0.005977
18	[jQAssistant, Rule, Concept]	19	0.006309
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006973
20	[Maven, Plugin]	21	0.006973
21	[Maven, Configuration]	21	0.006973
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007305
23	[Type, File, Java, ByteCode, Enum]	28	0.009297
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009961
25	[Xml, Namespace]	36	0.011953
26	[Type, File, Java, ByteCode, Annotation]	44	0.014610
27	[File, Directory]	44	0.014610
28	[Type, File, Java, Class, ByteCode, Throwable]	55	0.018262
29	[Type, File, Java, ByteCode, Interface, Generi...	85	0.028223

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

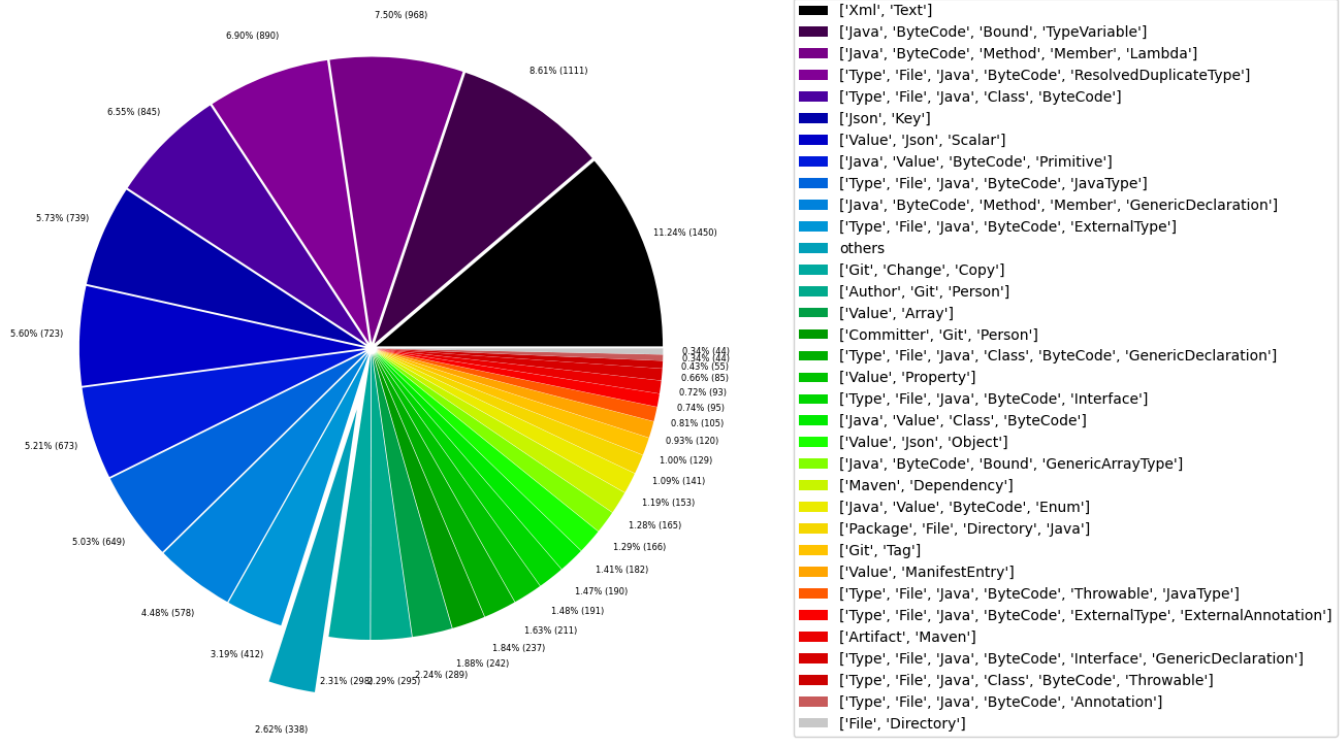


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

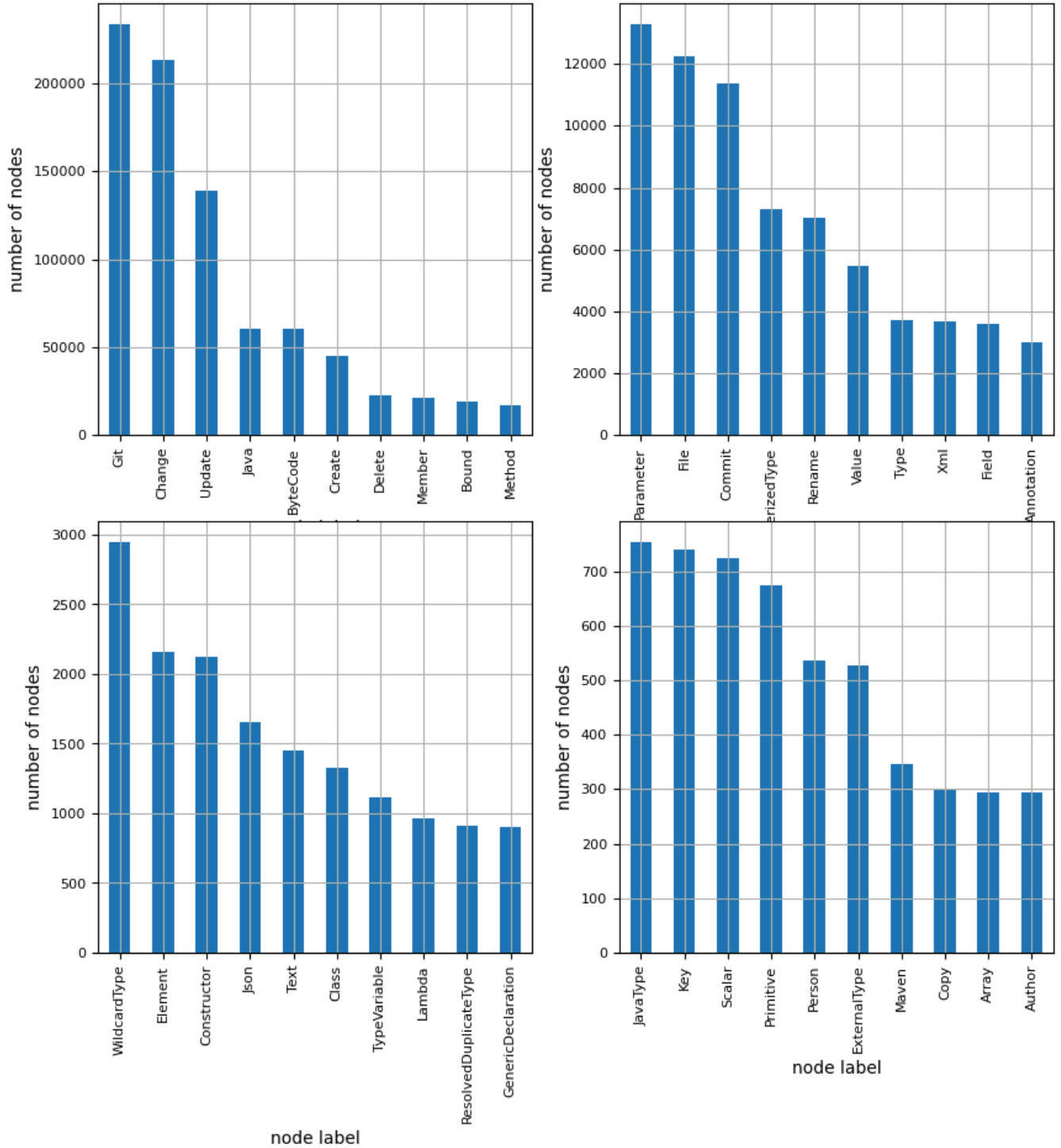
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.747120
1	Change	213807	70.992131
2	Update	138781	46.080619
3	Java	60668	20.144105
4	ByteCode	60478	20.081017
5	Create	44929	14.918153
6	Delete	22781	7.564166
7	Member	20744	6.887804
8	Bound	18758	6.228376
9	Method	17142	5.691802
10	Parameter	13306	4.418103
11	File	12244	4.065478
12	Commit	11390	3.781917
13	ParameterizedType	7286	2.419232
14	Rename	7018	2.330245
15	Value	5483	1.820566
16	Type	3718	1.234519
17	Xml	3675	1.220241
18	Field	3602	1.196002
19	Annotation	2980	0.989474
20	WildcardType	2950	0.979513
21	Element	2162	0.717867
22	Constructor	2124	0.705250
23	Json	1652	0.548527
24	Text	1450	0.481456
25	Class	1327	0.440615
26	TypeVariable	1111	0.368895
27	Lambda	968	0.321413
28	ResolvedDuplicateType	912	0.302819
29	GenericDeclaration	904	0.300163
30	JavaType	754	0.250357
31	Key	739	0.245376
32	Scalar	723	0.240064
33	Primitive	673	0.223462
34	Person	537	0.178305
35	ExternalType	528	0.175316
36	Maven	346	0.114885
37	Copy	298	0.098947
38	Array	295	0.097951
39	Author	295	0.097951

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 936089

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.840456
1	MODIFIES	213807	22.840456
2	UPDATES	138781	14.825620
3	CREATES	52245	5.581200
4	INVOKES	36805	3.931784
5	DELETES	29799	3.183351
6	COMMITTED	22780	2.433529
7	DEPENDS_ON	22504	2.404045
8	OF_TYPE	21945	2.344328
9	DECLARES	21204	2.265169
10	OF_RAW_TYPE	17367	1.855272
11	HAS	14444	1.543016
12	HAS_PARENT	13921	1.487145
13	RETURNS	12858	1.373587
14	HAS_COMMIT	11390	1.216765
15	READS	9381	1.002148
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.898846
17	HAS_FILE	8286	0.885172
18	RENAMES	7018	0.749715
19	OF_GENERIC_TYPE	6000	0.640965
20	RESOLVES_TO	4234	0.452307
21	SIMILAR	4078	0.435642
22	WRITES	3948	0.421755
23	CONTAINS	3928	0.419618
24	HAS_NEW_NAME	3662	0.391202
25	RETURNS_GENERIC	3597	0.384258
26	CHANGED_TOGETHER_WITH	3034	0.324114
27	ANNOTATED_BY	2924	0.312363
28	REQUIRES	2233	0.238546
29	HAS_FIRST_CHILD	2162	0.230961

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

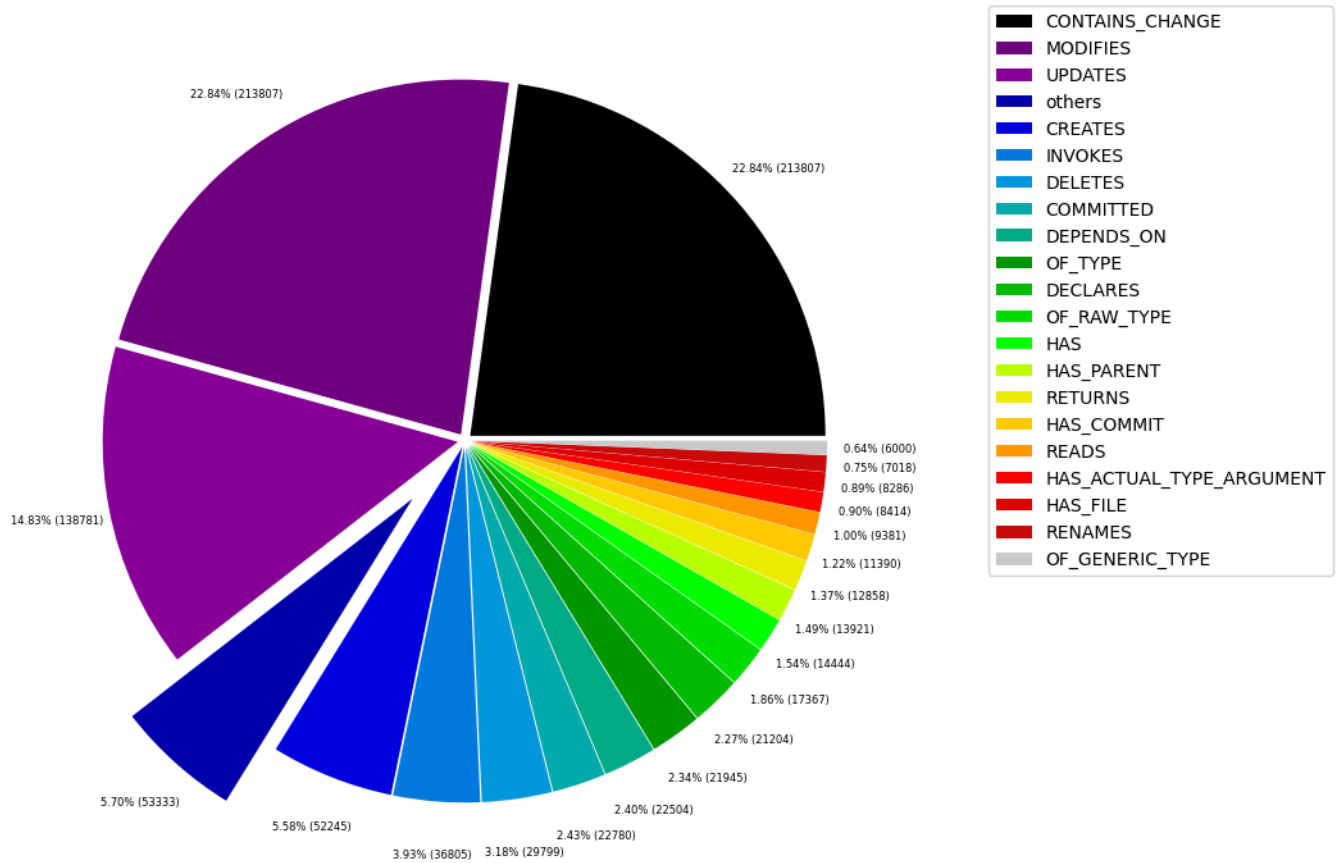


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

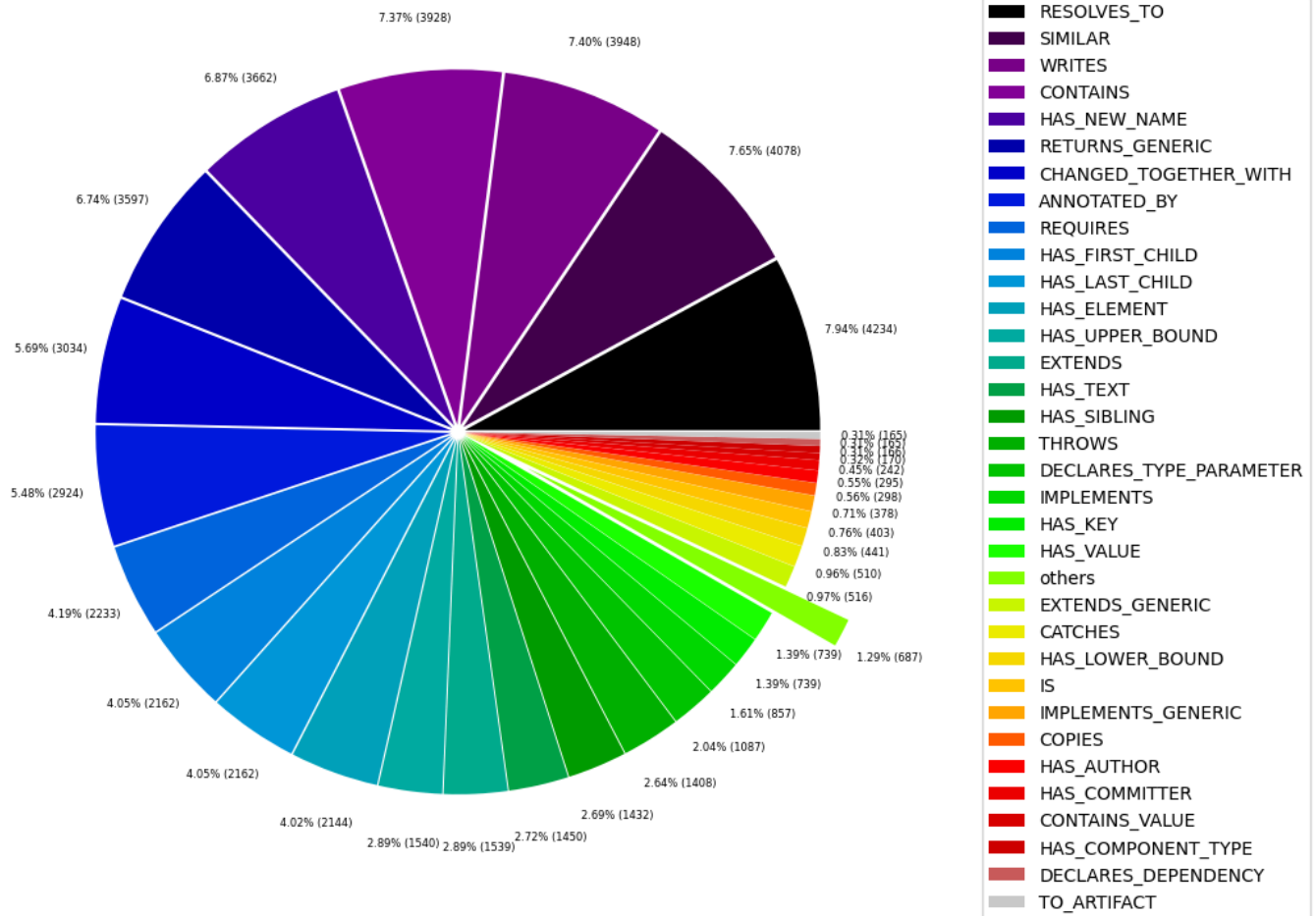
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000961
6	HAS_ROOT_ELEMENT	10	0.001068
7	HAS_GOAL	16	0.001709
8	HAS_EXECUTION	16	0.001709
9	OF_NAMESPACE	18	0.001923
10	HAS_ATTRIBUTE	18	0.001923
11	INCLUDES_CONCEPT	19	0.002030
12	USES_PLUGIN	21	0.002243
13	IS_ARTIFACT	21	0.002243
14	HAS_CONFIGURATION	21	0.002243
15	REQUIRES_TYPE_PARAMETER	24	0.002564
16	REQUIRES_CONCEPT	28	0.002991
17	DECLARES_NAMESPACE	36	0.003846
18	HAS_DEFAULT	39	0.004166
19	COPY_OF	119	0.012712
20	HAS_TAG	129	0.013781
21	ON_COMMIT	129	0.013781
22	DECLARES_DEPENDENCY	165	0.017627
23	TO_ARTIFACT	165	0.017627
24	HAS_COMPONENT_TYPE	166	0.017733
25	CONTAINS_VALUE	170	0.018161
26	HAS_COMMITTER	242	0.025852
27	HAS_AUTHOR	295	0.031514
28	COPIES	298	0.031835
29	IMPLEMENTS_GENERIC	378	0.040381

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	UPDATES	[File, Git]	138781	138781	
1	[Git, Update, Change]	MODIFIES	[File, Git]	138781	138781	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	138781	11390	
3	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
4	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22302	13369	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8480	13369	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8459	13369	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3466	7245	
25	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3114	7286	
26	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2698	13306	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2520	2120	
29	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301170
total_number_of_relationships (edges): 936089
-> total directed graph density: 1.0320367584330215e-05
-> total directed graph density in percent: 0.0010320367584330215
```